

STAT 4037 Lab1 R

2023-01-27

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Reading in the data set using `file.choose` and `read.csv` because I kept getting errors about empty columns using `read.table` and when I attempted to pass the `fill=TRUE` argument into the command it botched the data set.

```
file.choose()
```

```
## [1] "C:\\Users\\super\\OneDrive\\Documents\\STAT4307_Lab1_Rknight_RollerK.Rmd"
```

```
drugs<-read.csv("C:\\Users\\super\\Downloads\\drugsurv.csv", header=TRUE)
```

```
head(drugs)
```

```
##   Gender Age Education Ever.Smoker Smoking.Status Mariguana Alcohol.Dep.
## 1 FEMALE  20         14         NO         never         NO         NO
## 2  MALE   36         12        YES        current        YES         NO
## 3 FEMALE  83         10         NO         never         NO         NO
## 4  MALE   41         13        YES        former        YES         NO
## 5 FEMALE  71         17        YES        former         NO         NO
## 6  MALE   31         12        YES        current        YES         NO
##   Age.Group
## 1      <25
## 2    25-39
## 3     65+
## 4    40-64
## 5     65+
## 6    25-39
```

```
attach(drugs)
```

```
names(drugs)
```

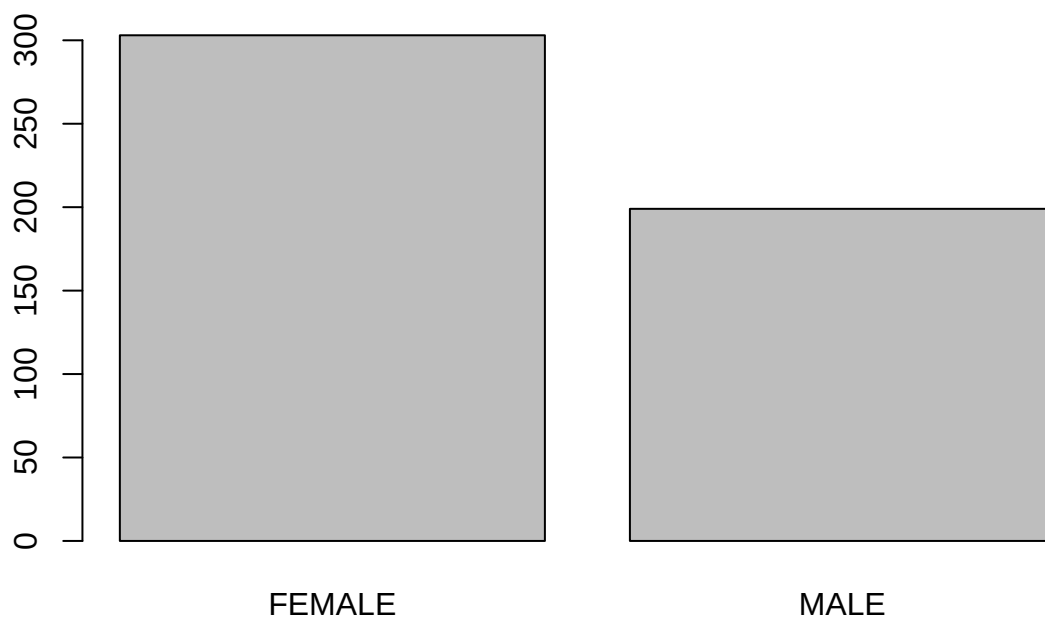
```
## [1] "Gender"      "Age"         "Education"    "Ever.Smoker"
## [5] "Smoking.Status" "Mariguana"    "Alcohol.Dep." "Age.Group"
```

Table and Graph for Gender

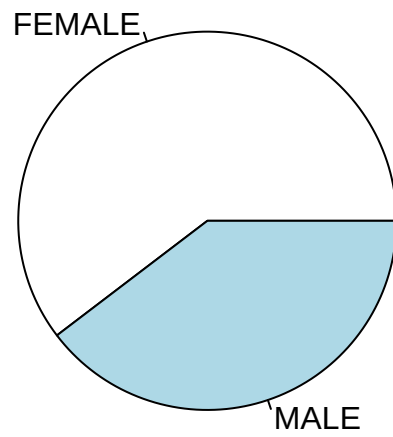
```
table(Gender)
```

```
## Gender  
## FEMALE MALE  
##    303   199
```

```
counts<-table(Gender)  
barplot(counts)
```



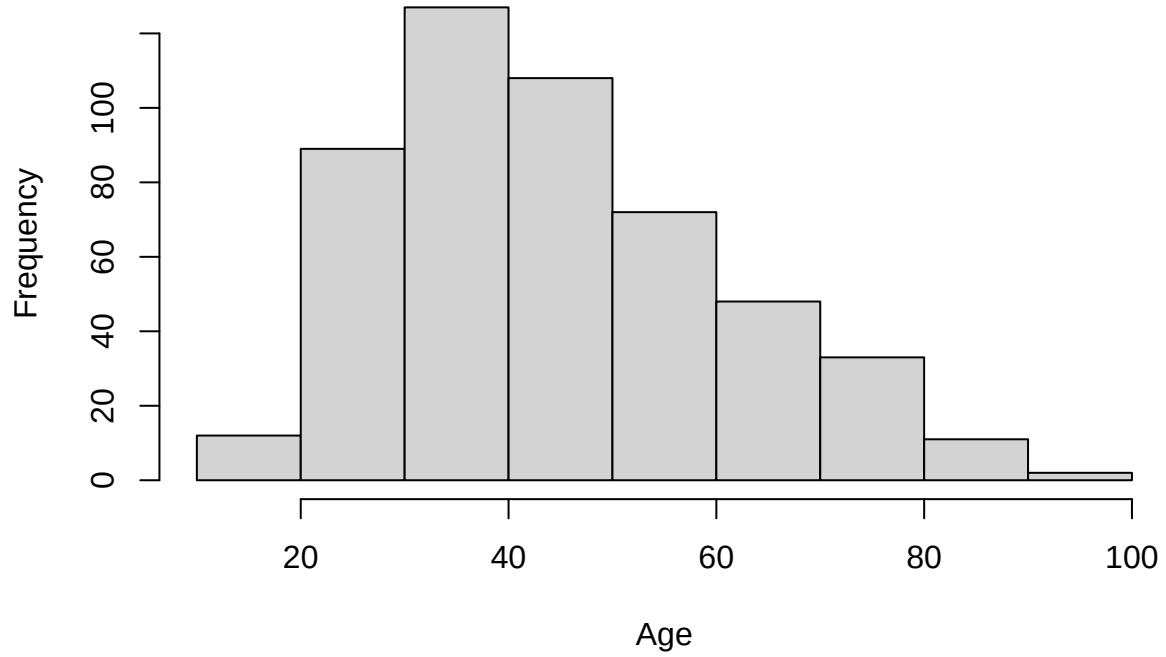
```
pie(counts)
```



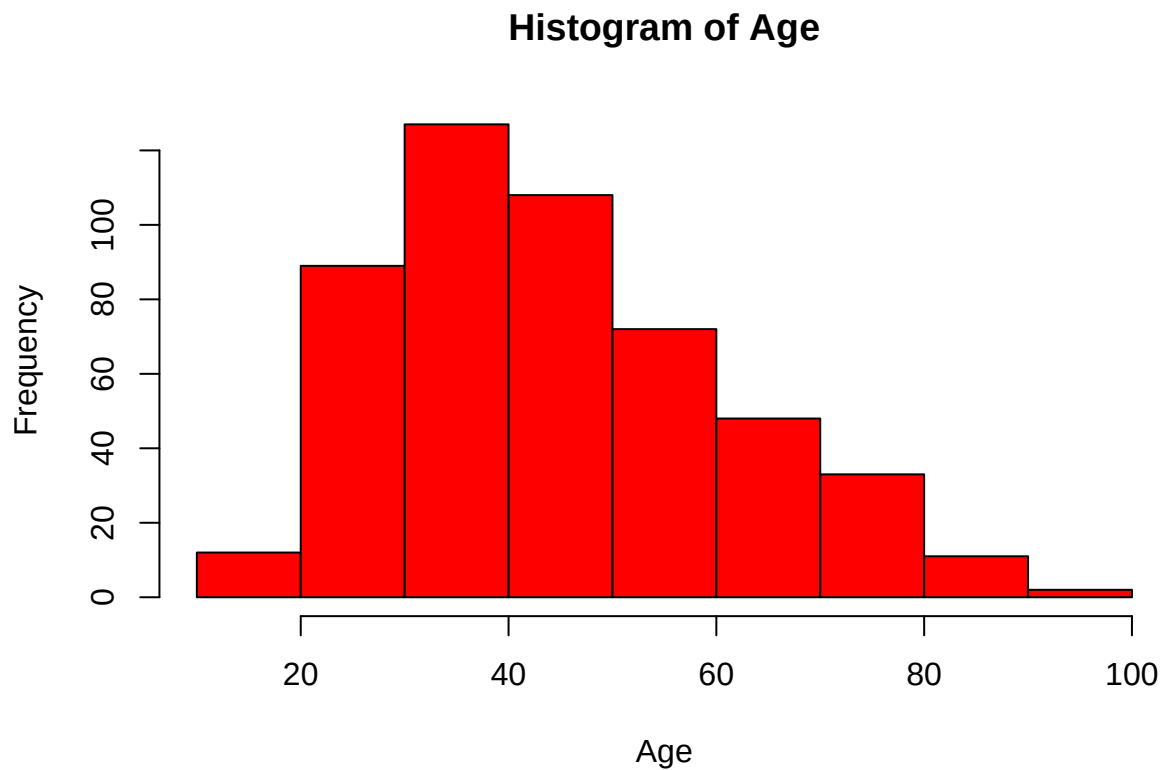
Creating a Histogram for age

```
hist(Age)
```

Histogram of Age



```
hist(Age, col="red")
```



Recoding age in three groups:

If there are only two classes to use:

```
agenew<-ifelse(Age<50, "young", "old")
table(agenew)
```

```
## agenew
##   old young
##   177   325
```

If there are **THREE** groups

```
agegroup3<-ifelse(Age<40, "young", ifelse(Age>40 & Age<60, "middle aged", "old"))
table(agegroup3)
```

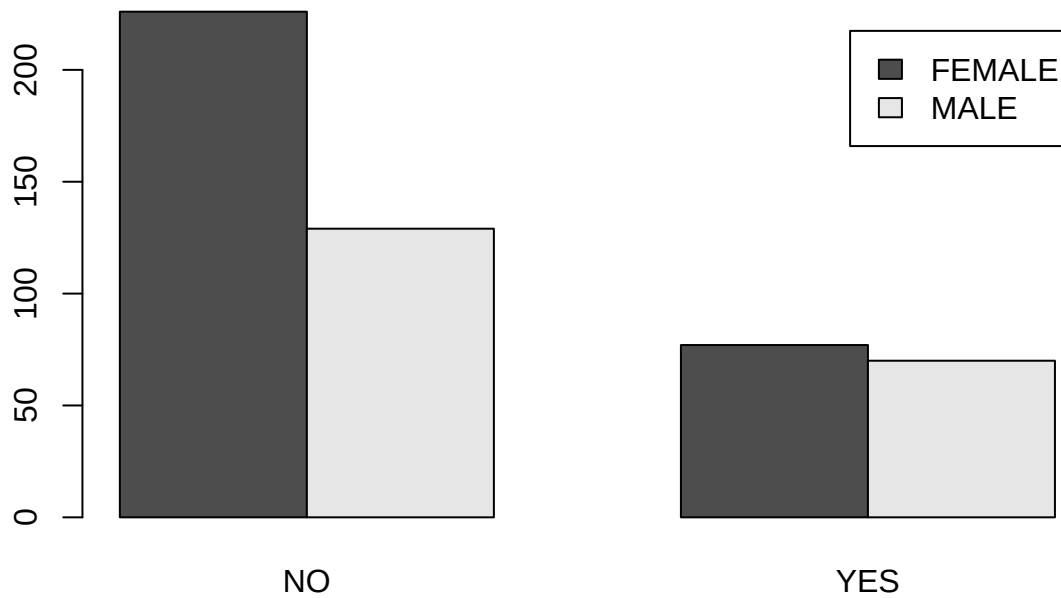
```
## agegroup3
## middle aged      old      young
##           171      117      214
```

Creating a new Quantitative Variable

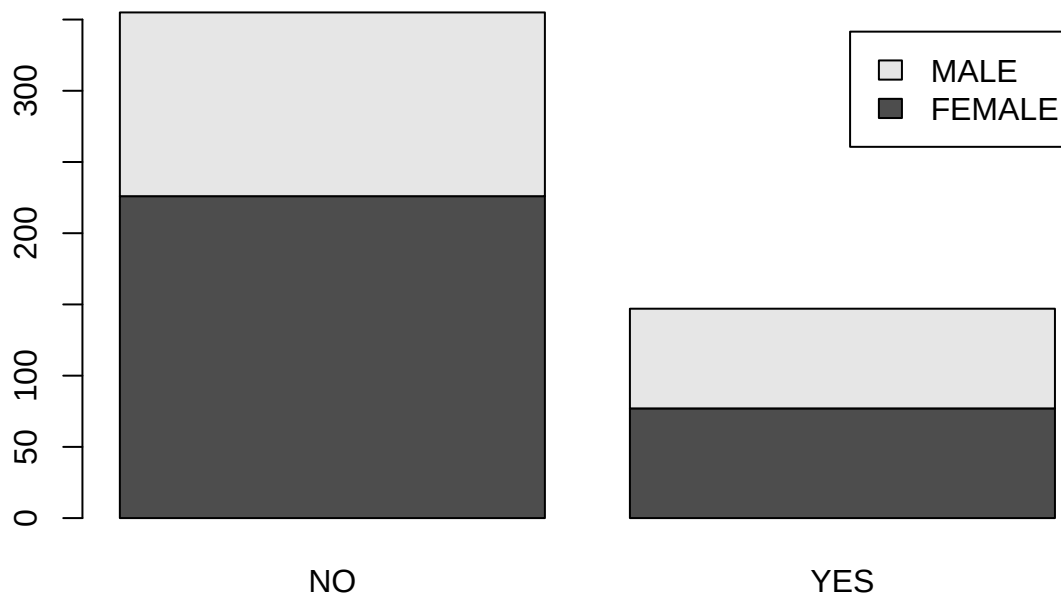
```
agenow<-Age+15
```

Cross Tabulation of gender and marijuana

```
counts<-table(Gender,Mariguana)
barplot(counts, beside=TRUE, legend=rownames(counts))
```



```
barplot(counts, beside=FALSE, legend=rownames(counts))
```



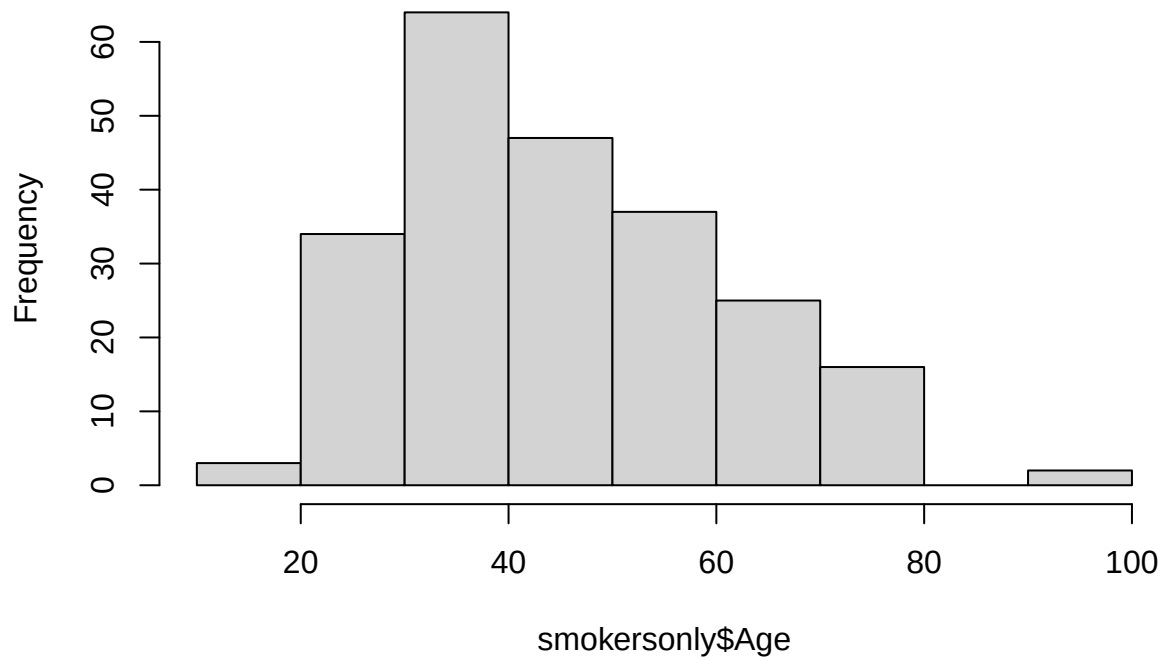
Creating a subset formed just by the ever smokers

```
smokersonly<-drugs[Ever.Smoker=="YES",]
head(smokersonly)
```

```
##   Gender Age Education Ever.Smoker Smoking.Status Mariguana Alcohol.Dep.
## 2   MALE  36       12        YES      current      YES         NO
## 4   MALE  41       13        YES      former      YES         NO
## 5 FEMALE  71       17        YES      former      NO          NO
## 6   MALE  31       12        YES      current      YES         NO
## 7   MALE  60       18        YES      former      YES         NO
## 8   MALE  41       12        YES      former      NO          NO
##   Age.Group
## 2   25-39
## 4   40-64
## 5   65+
## 6   25-39
## 7   40-64
## 8   40-64
```

```
hist(smokersonly$Age)
```

Histogram of smokersonly\$Age

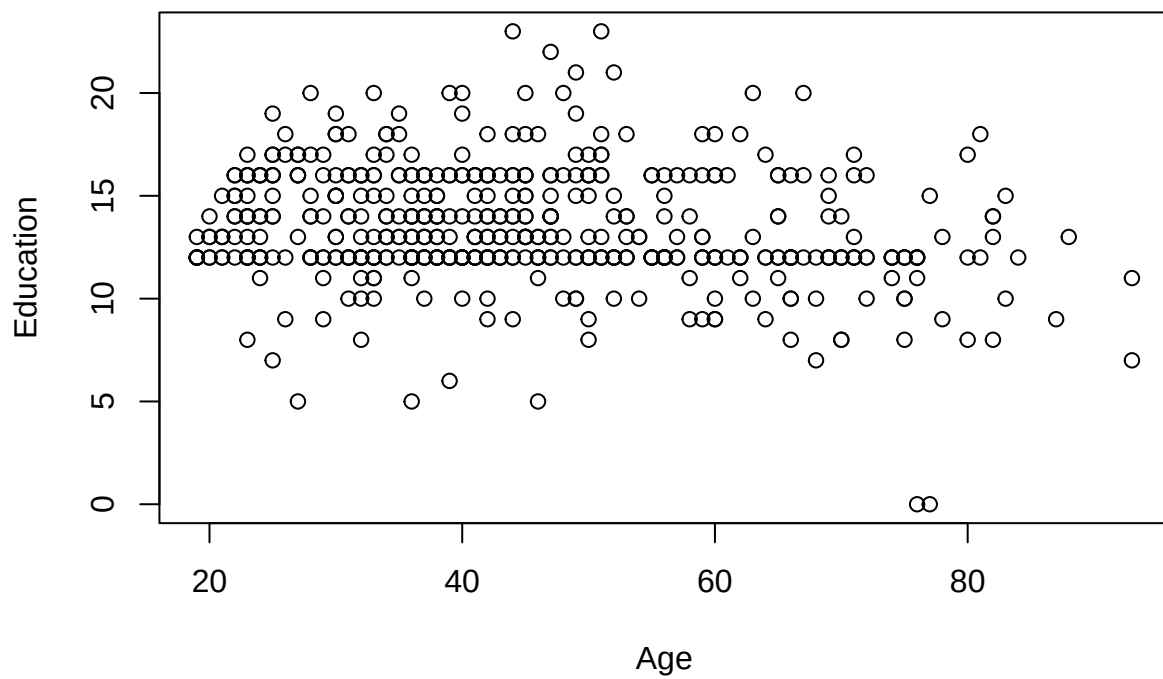


Calculating correlation between age and years of education

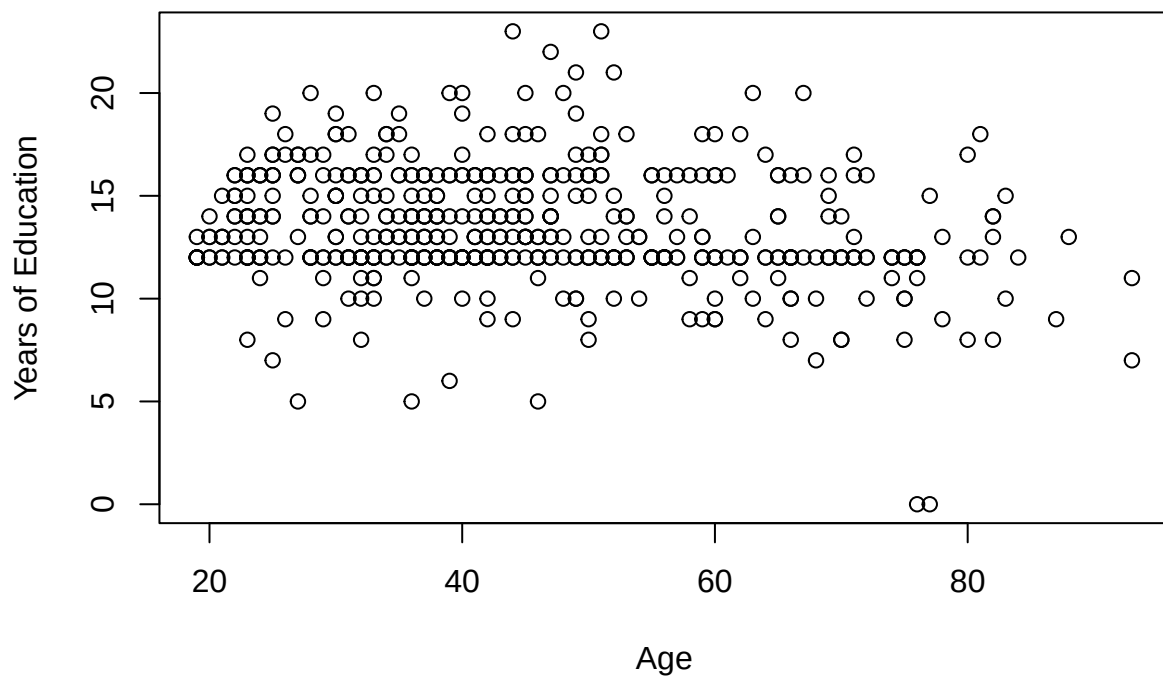
```
cor(Age,Education)
```

```
## [1] -0.2027511
```

```
plot(Age,Education)
```



```
plot(Age, Education, xlab="Age", ylab="Years of Education")
```



More things to do with R

```
mean(Age)
```

```
## [1] 45.06375
```

```
var(Age)
```

```
## [1] 270.2275
```

```
summary(Age)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      19.00   33.00   42.00   45.06   56.00   93.00
```

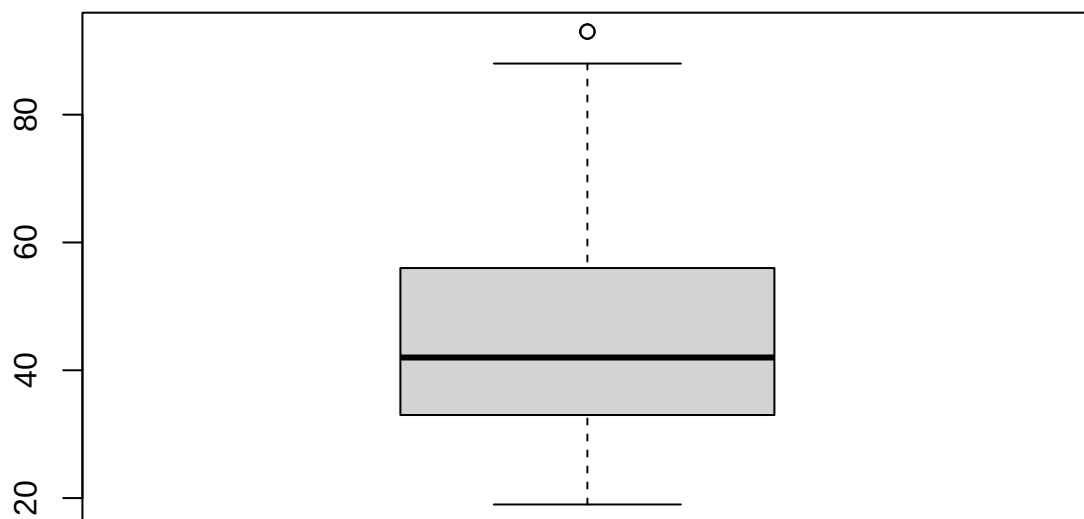
```
sd(Age)
```

```
## [1] 16.4386
```

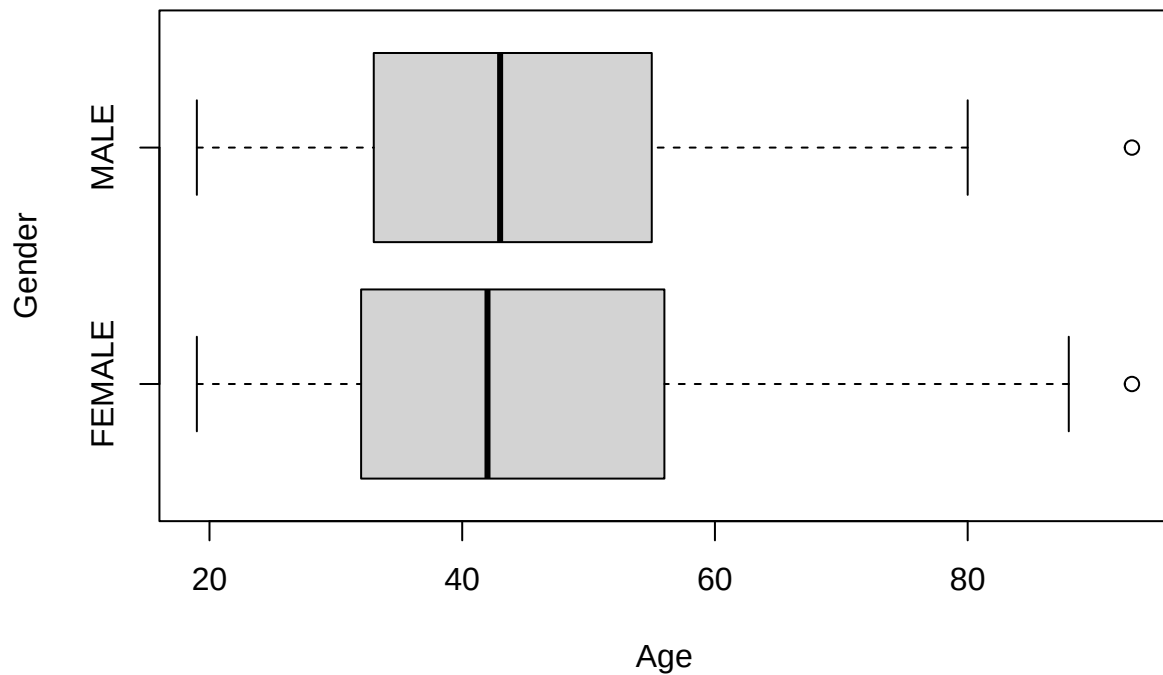
```
tapply(Age, Gender, summary)
```

```
## $FEMALE
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   19.00   32.00   42.00   44.93   56.00   93.00
##
## $MALE
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   19.00   33.00   43.00   45.26   55.00   93.00
```

```
boxplot(Age)
```



```
boxplot(Age~Gender, horizontal=TRUE, xlab="Age")
```



```
mytable<-table(Gender,Smoking.Status)
mytable
```

```
##           Smoking.Status
## Gender   current former never
##  FEMALE      68     47    188
##   MALE       53     63     83
```

```
margin.table(mytable,1)
```

```
## Gender
## FEMALE  MALE
##   303   199
```

```
margin.table(mytable,2)
```

```
## Smoking.Status
## current  former  never
##    121    110    271
```

```
prop.table(mytable,1)
```

```
##           Smoking.Status
```

```
## Gender      current    former    never
##   FEMALE 0.2244224 0.1551155 0.6204620
##   MALE   0.2663317 0.3165829 0.4170854
```

```
prop.table(mytable,2)
```

```
##           Smoking.Status
## Gender      current    former    never
##   FEMALE 0.5619835 0.4272727 0.6937269
##   MALE   0.4380165 0.5727273 0.3062731
```